

```

function mobMap_denseMatching_(images, images_d, vSet_gt, N, K, base)

%% p9 && P11: dense matching

% ==> rectify fehlt noch

figure(2);
for viewId = 1 : N

    %% P8: load left images and disp map
    Irgb = readimage(images, viewId);
    dispMap = readimage(images_d, viewId);

    %% P8: reconstruct scene
    xyzPoints_stereopair = mobMap_denseMatching_helper(dispMap, K, Irgb, 300, base);
    pc{viewId}=xyzPoints_stereopair;

    %% P11: transform to world
    pc_tmp = pc{viewId};
    XYZ_w = [ ];
    Color_w = [ ];
    Tl2w = vSet_gt.poses.Location{viewId};
    Rl2w = vSet_gt.poses.Orientation{viewId}'; % Matlab Notation ==> 'normal'

    for i=1 : length(pc_tmp.Location)
        XYZ_l = pc_tmp.Location(i, 1:3);
        P = Rl2w*XYZ_l' + Tl2w';
        XYZ_w(i,1:3) = P';
    end

    pc_tmp = pointCloud(XYZ_w, 'Color', pc_tmp.Color);
    if viewId==1
        Pc_all =pc{1};
    else

```

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        Pc_all = pcmerge(Pc_all, pc_tmp, 0.5);
    end

    %% P8: plot
    pcshow(Pc_all, 'MarkerSize',50, 'VerticalAxis', 'Y', 'VerticalAxisDir', 'Down');
    title('3D modell multi view geometrie');
    axis equal;
    xlabel('X (cm)');
    ylabel('Y (cm)');
    zlabel('Z (cm)');
    xlim([-125, 125]);
    ylim([-125, 125]);
    zlim([50, 300]);
    grid on;

end

```